Minghan Lyu

Home page | Email: 202300300382@mail.sdu.edu.cn

EDUCATION

Shandong University (SDU), Jinan, China Sep 2023 - Present

Bachelor of Engineering in Software Engineering (Emerging Engineering Honors Program in Artificial Intelligence), in progress at the School of Software. The program is offered under the Joint SDU-NTU Centre for Artificial Intelligence Research (C-FAIR).

Related courses: Numerical Computing (A+), Information Retrieval (A+), Linear Algebra (A), Computer Organization and Architecture (A), Introduction to the Artificial Intelligence Major (A+), Social Networks and Recommender Systems (A)

RESEARCH EXPERIENCE

Machine Learning for Protein Modeling Jun 2026 - Sep 2026

Advisor: Sorin Istrail (Department of Computer Science, Brown University; Professor)

- Led a three-member course project on the transferability of machine-learning protein models; developed the CGSchNet analysis and integrated the final report.
- Developed a model that predicts persistent molecular contacts from static structures to improve estimates of protein flexibility.
- Improved protein-flexibility prediction (RMSF Spearman correlation) by 2.7% over a uniform Gaussian network model; validated robustness through resampling and ablation studies.

Policy-Guided Mango Shape Annealing Apr 2026 - Aug 2026

Advisor: John H. Reif (Trinity College of Arts and Sciences, Duke University; Professor)

- Guided DNA origami shape design with feasibility-aware proposal selection to address inefficient stochastic search.
- Designed constraint-aware actions that exclude invalid combinations of design rules, sites, and parameters.
- Improved mean design objective scores by 5.8% over Mango-ShapeAnneal across DNA origami search benchmarks while preserving manufacturability constraints. GitHub

Eukaryotic Gene Structure Prediction with HMM Oct 2025 - Jan 2026

Advisor: Jun Wang (School of Software, Shandong University; Professor)

- Adapted repeat-aware decoding to a multi-state HMM, combining coding-sequence phase modeling with repeat states and softmask penalties to address false-positive exons.
- Diagnosed the failure of naive repeat encoding and evaluated alternative decoders through ablation studies, achieving balanced nucleotide-level sensitivity and precision on held-out genomic regions. GitHub

Traffic Forecasting on METR-LA Sep 2025 - Dec 2025

Advisor: Yongshun Gong (School of Software, Shandong University; Professor)

- Addressed missing observations and changing traffic patterns with a graph-based model combining spatial relationships and temporal learning.
- Reduced test mean absolute error (MAE) by 6.3% on METR-LA through spatial-temporal modeling and regional adaptation. GitHub

News Recommendation with NPA Jan 2025 - Jun 2025

Advisor: Li Lian (School of Software, Shandong University; Associate Professor)

- Introduced pretrained DistilBERT representations of news titles and abstracts into an NPA-based recommender to extend its original title-only modeling.
- Observed consistent ranking gains over the original NPA model on MIND; used controlled negative sampling and automated hyperparameter tuning for systematic evaluation.

LLM-based Paper Screening Nov 2024 - Dec 2024

Advisor: Yuqing Sun (School of Software, Shandong University; Professor)

- Automated title and abstract screening for systematic reviews using LLM prompts aligned with inclusion and exclusion criteria.
- Added reproducible decision logs and lightweight checks to support faster screening and consistent review decisions.

PROJECT EXPERIENCE

Yantu Lingban: Agentic Exam-Preparation Companion System Mar 2026 - Jun 2026

Advisor: Ning Liu (School of Software, Shandong University; Assistant Researcher)

- Built a Windows companion that unifies exam study planning, question answering, and mistake review in one adaptive learning workflow.
- Coordinated specialized study and well-being agents through a shared user-state model to personalize assistance.
- Enabled proactive reminders, spaced review, and plan adjustments based on learning progress; integrated screenshot question answering. Code available on Gitee.

Multimodal Understanding for Product Live-stream Sep 2025 - Dec 2025

Advisor: Lei Meng (School of Software, Shandong University; Professor)

- Combined video frames, speech, and product metadata to retrieve products from noisy live streams.
- Used hard-negative contrastive learning and product-region localization to distinguish similar items and reduce background noise.
- Improved cross-modal product similarity scores by 7.4% through hard-negative mining and product-region localization. GitHub

Brand-Geographic Grid Prediction Mar 2025 - Jul 2025

Advisor: Meng Chen (School of Software, Shandong University; Associate Professor)

- Developed a location-ranking model to support brand site selection from sparse trajectories with limited timing information.
- Combined trajectory patterns with geographic context, using density-aware ordering to model spatial relationships.
- Improved first-choice location prediction accuracy (Acc@1) by 3% over the best LSTM-MLP baseline for brand site selection.

INTERNSHIP

Invoice Reimbursement Review System | Advisor: Siwei Chen Jul 2026

Intern | Research and Practical Teaching Center, Chinese Aeronautical Establishment

- Built a reimbursement review system that connects student submissions with teacher review, automating document extraction and invoice-level reconciliation.
- Combined document parsing, OCR, and optional vision-language models with manual review and Excel/PDF reporting; packaged the system for Windows deployment.
- Assisted Coze-based AI programming labs for Khalifa University exchange students.

PUBLICATION

Lyu, M. (2026). Predicting Molecular Dynamics Contact Persistence from Static Structures for Weighted Gaussian Network Models. Theoretical and Natural Science, 190, 142-154.

ICBioMed 2026 symposium: Biomedical Engineering and Smart Medicine (ICBM_HZ 2026).

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AWARDS AND HONORS

- Mathematical Modeling National Contest, Problem C, Provincial First Prize (8%) 2 times in Year 2024 and 2025
- Innovation Research Specialty Scholarship 2024-2025 Academic Year
- 2026 COMAP Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM), Meritorious Winner (M Award) (6.67%)

SKILLS AND OTHERS

Programming language: C/C++, Python, Java, MySQL, MATLAB, LaTeX, R

Framework & Tools: PyTorch, GitHub, TensorFlow, Java IDE, Spring Web, Thymeleaf, Hugging Face, ROS, Ubuntu, Anaconda, BLAST, MEME Suite, RStudio, Git